

Topic Covered:

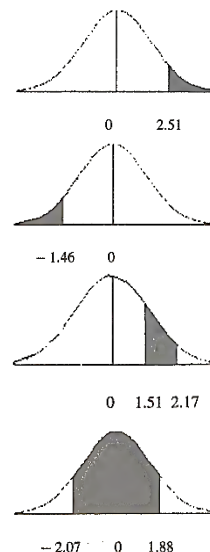
- Normal distribution
- Application of normal distribution
- Central limit theorem

1. Find the probabilities for each, using the standard normal distribution.

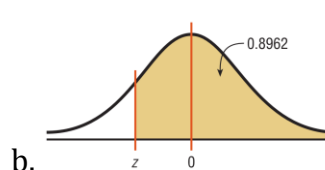
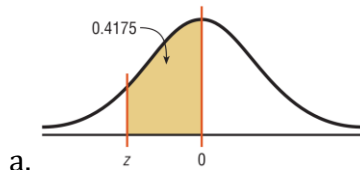
- a. $P(z > 2.51)$
- b. $P(z < -1.46)$
- c. $P(1.51 < z < 2.17)$
- d. $P(-2.07 < z < 1.88)$

Answer:

- a. $P(z > 2.51) = 1 - P(z < 2.51) = 1 - \phi(2.51) = 1 - 0.9940 = 0.0060$
- b. $P(z < -1.46) = 1 - P(z < 1.46) = 1 - \phi(1.46) = 1 - 0.9279 = 0.0721$
- c. $P(1.51 < z < 2.17) = \phi(2.17) - \phi(1.51) = 0.9850 - 0.9345 = 0.0505$
- d. $P(-2.07 < z < 1.88) = \phi(1.88) - \phi(-2.07) = \phi(1.88) - [1 - \phi(2.07)]$
 $= \phi(1.88) + \phi(2.07) - 1$
 $= 0.9699 + 0.9808 - 1 = 0.9507$



2. Find the z value that corresponds to the given area.



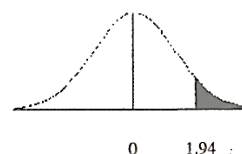
Answer:

- a) Since the z score is on the left side of 0, use the negative z table. Areas in the negative z table are in the tail, so we will use $0.5000 - 0.4175 = 0.0825$ as the area. The closest z score corresponding to an area of 0.0825 is $z = -1.39$. [i.e., $P(z < -1.39) = 0.0825$].
- b) Use the negative z table and $1 - 0.8962 = 0.1038$ for the area. The z score is $z = -1.26$. [i.e., $P(z < -1.26) = 0.1038$].

3. Miles Driven Annually: The mean number of miles driven per vehicle annually in the United States is 12,494 miles. Choose a randomly selected vehicle, and assume the annual mileage is normally distributed with a standard deviation of 1290 miles. What is the probability that the vehicle was driven more than 15,000 miles? Less than 8000 miles? Would you buy a vehicle if you had been told that it had been driven less than 6000 miles in the past year?

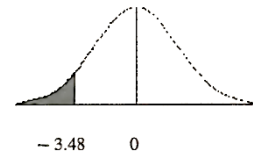
Answer:

For $x > 15,000$ miles: $z = \frac{x - \mu}{\sigma} = \frac{15,000 - 12,494}{1290} = 1.94$
 $\Rightarrow P(z > 1.94) = 1 - P(z < 1.94) = 1 - 0.9738 = 0.0262$



$$\# \text{ For } x < 8000 \text{ miles: } z = \frac{x - \mu}{\sigma} = \frac{8000 - 12.494}{1290} = -3.48$$

$$\Rightarrow P(z < -3.48) = 1 - P(z < 3.48) = 1 - 0.9997 = 0.0003$$



$$\# \text{ For } x < 6000 \text{ miles: } z = \frac{x - \mu}{\sigma} = \frac{6000 - 12.494}{1290} = -5.03$$

$$\Rightarrow P(z < -5.03) = 1 - P(z < 5.03) = 1 - 0.9999 = 0.0001$$

Maybe it would be good to know why it had only been driven less than 6000 miles.

4. New Home Sizes: A contractor decided to build homes that will include the middle 80% of the market. If the average size of homes built is 1810 square feet, find the maximum and minimum sizes of the homes the contractor should build. Assume that the standard deviation is 92 square feet and the variable is normally distributed.

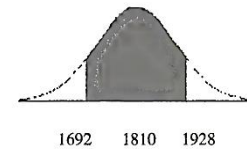
Answer:

The middle 80% leaves 10% in each tail, so we find the z-score where the cumulative area is 0.90 (or 0.10 on the left). The corresponding z scores will be ± 1.28 .

Since, $z = \frac{x - \mu}{\sigma} \Rightarrow x = z\sigma + \mu$. Then

$$x = +1.28(92) + 1810 = 1927.76 \text{ sq. ft. (maximum)}$$

$$x = -1.28(92) + 1810 = 1692.24 \text{ sq. ft. (minimum)}$$



5. The marks of 300 students of SUT in the mathematics examination are normally distributed with a mean of 172 marks and a standard deviation of 8 marks. Determine how many students have marks:
- between 164 and 180 marks.
 - Less than 164 marks.

Answer:

$$(a) z_1 = \frac{164 - 172}{8} = \frac{-8}{8} = -1.00 \quad \text{and} \quad z_2 = \frac{180 - 172}{8} = \frac{8}{8} = 1.00$$

$$\Rightarrow P(164 < X < 180) = P(-1.00 < Z < 1.00) = P(Z < 1.00) - P(Z < -1.00) =$$

$$= P(Z < 1.00) - [1 - P(Z < 1.00)] = 2 P(Z < 1.00) - 1 =$$

$$= 2(0.8413) - 1 = 0.6826$$

$$\Rightarrow \text{Number of students: } 300 \times 0.6826 = 204.78 \approx 205 \text{ students}$$

$$(b) z = \frac{164 - 172}{8} = \frac{-8}{8} = -1.00$$

$$\Rightarrow P(X < 164) = P(Z < -1.00) = 1 - P(Z < 1.00) = 1 - 0.8413 = 0.1587$$

$$\Rightarrow \text{Number of students: } 300 \times 0.1587 = 47.61 \approx 48 \text{ students}$$

6. Water Use: The Old Farmer's Almanac reports that the average person uses 123 gallons of water daily. If the standard deviation is 21 gallons, find the probability that the mean of a randomly selected sample of 15 people will be between 120 and 126 gallons. Assume the variable is normally distributed.

Answer:

Applying the central limit theorem:

$$\# z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{120 - 123}{\frac{21}{\sqrt{15}}} = -0.55$$

$$\# z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{126 - 123}{\frac{21}{\sqrt{15}}} = 0.55$$

$$\Rightarrow P(-0.55 < z < 0.55) = 0.7088 - 0.2912 = 0.4176 \quad (\text{or } 41.76\%)$$

